CYBER SECURITY OPERATIONS - Safa

Exam Study Guide

Week 2: IT Systems Landscape + Service Mesh

1. The 7-Phase Cyber Security Operations Process Cycle

The process cycle maps each phase to a NIST 2.0 function. The cycle is continuous — Phase 07 feeds back into remediation and resilience across all phases.

Phase	Name	Key Function	NIST Mapping
01	IT Systems Landscape	Identify & document entire IT environment	Identify
02	Aggregation & Normalisation	Consolidate & standardise security data	Protect
03	SOC & SIEM Functions	Monitor, analyse, correlate events; identify IoCs	Protect + Detect
04	Service Management Integration	Align security with operational objectives	Govern
05	Workflow Automation	Streamline repetitive tasks & incident response	Protect + Detect + Respond
06	Incident Response	Contain, mitigate, remediate identified threats	Respond
07	Remediation, Recovery, Resilience	Restore systems; prevent recurrence	Recover

Remember: Remediation → Recovery → Resilience runs as a continuous loop above all phases, not just Phase 07.

2. Phase 01 — IT Systems Landscape (Your Main Focus)

What it does

Identifies and documents the organisation's full IT environment: structure, components, architecture, systems, tools, processes, and objectives. Security operations cannot begin without understanding what exists.

The 5 IT domains

Domain	Key Components
Infrastructure & Network	Global WAN, multi-cloud (AWS/Azure/GCP), SDN, load balancers, CDN
Communication & Collaboration	Unified comms, global email, document management, virtual meetings
Security & Compliance	IAM, MFA, SIEM, DLP, SOC, threat intelligence, vulnerability management, incident response
Data Management	Distributed databases, data warehousing, log aggregation, BI, real-time streaming
Development & Operations	CI/CD, container orchestration, microservices, API gateway, service mesh, monitoring

Integration task (exam context)

- **Primary domain:** Where does your project sit in this landscape?
- **Data IN:** What data does your project receive from other systems?
- **Data OUT:** What alerts/data does your project push to other systems?
- **Glossary:** Two unfamiliar concepts to research and add to the shared glossary.

Glossary contribution template

```
Term | Definition (own words) | Relevance to project | Context in
landscape | Source | Added by
```

3. Service Mesh

Definition

A dedicated infrastructure layer that handles service-to-service communication in a microservices architecture. It manages traffic routing, security (mTLS), observability, and fault tolerance WITHOUT changing application code.

Architecture

- **Sidecar proxy —** Deployed alongside each microservice; intercepts ALL inbound/outbound traffic on the service's behalf
- **Data plane —** The collection of all sidecar proxies; handles actual traffic
- **Control plane —** Distributes routing rules, manages TLS certificates, collects telemetry metrics
- **Istio —** Most widely used open-source service mesh; uses Envoy as its sidecar proxy

Benefits (exam-likely points)

- **Security —** mTLS encrypts all inter-service traffic; adds auth without touching code
- **Policy enforcement —** Rate limiting, auth policies, quotas enforced centrally at proxy level
- **Traffic management —** Canary deployments, load balancing, version-based routing
- **Observability —** Distributed tracing, metrics (latency, error rates), health checks
- **Fault tolerance —** Auto retries, rerouting on failure — apps stay up when a service goes down

Challenges

- Complex to set up; harder in multi-cloud or legacy environments
- Each sidecar increases CPU and memory usage
- Teams need training; mTLS misconfigs are hard to debug

Service mesh vs API gateway

	Service Mesh	API Gateway
Traffic type	Internal (service-to-service)	External (client-to-backend)
Position	Inside the cluster/mesh	Edge of the system

Used together?	Yes — mesh secures internal; gateway handles external	

4. Quick-Reference Facts

- **Scenario org:** 8 offices, 6 time zones, 3,000 employees — the scenario org size
- **SDN:** SDN = Software-Defined Networking (optimal routing layer)
- **IoC:** IoC = Indicator of Compromise (detected in Phase 03 via SIEM)
- **SIEM:** SIEM = Security Information and Event Management (correlates events, Phase 03)
- **DLP:** DLP = Data Loss Prevention
- **IAM:** IAM = Identity & Access Management
- **mTLS:** mTLS = Mutual TLS (both sides authenticate)
- **Envoy:** Envoy = the proxy Istio uses in its data plane